

MD. FERDAOUS AL-FARABE

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CAREER OBJECTIVE

Industrial and Production Engineering graduate focused on Operations Research, particularly multi-objective and combinatorial optimization using evolutionary algorithms and mathematical programming, together with machine learning. I have applied these methods to scheduling and supply chain problems, and my current work extends into reinforcement learning. My goal is to develop scalable, data-driven optimization frameworks for practical manufacturing and supply chain systems.

EDUCATION

Rajshahi University of Engineering & Technology (RUET)

Rajshahi, Bangladesh

B.Sc. Engineering in Industrial & Production Engineering

Mar 2022 – Aug 2026

CGPA: **3.53/4.00**

- **Thesis:** Criticality-Guided Memetic NSGA-II for University Course Timetabling: A Multi-Objective Flexible Job Shop Scheduling Approach. Developed a multi-objective optimization model for university course scheduling, balancing timetable efficiency, workload distribution, and resource utilization.
- **Case Study:** A Bi-Objective Optimization Model for a Pharmaceutical Company Distribution Network (Bangladesh). Built a MILP model minimizing distribution cost and maximizing job creation across 64 districts, validated on a leading pharmaceutical company's anti-diabetic drug distribution network.
- **Notable Courses:** Operations Research, Supply Chain Management, Operations Management, Probability & Statistics, Engineering Economy, Industrial Simulation Lab, Numerical Analysis Lab.

RESEARCH INTERESTS

- Operations Research
- Machine Learning and Data Science
- Logistics Network Design
- Mathematical Modeling and Optimization
- Artificial Neural Networks (ANN)
- Supply Chain Management

PUBLICATIONS

Journal Articles

Under Review

[R-1] Rupok Islam Avi, **Md. Ferdaous Al-Farabe**, Khairun Nahar. “Criticality-Guided Memetic NSGA-II for University Course Timetabling: A Multi-Objective Flexible Job Shop Scheduling Approach.” *Computers & Operations Research*, Elsevier. [\[Paper\]](#)

Conference Publications

Published

[C-1] Rupok Islam Avi, **Md. Ferdaous Al-Farabe**, Ifte Kharul Islam Nahid, Md. Limonur Rahman Lingkon, Md. Sanowar Hossain. “Integrating Demand Forecasting and EOQ for Inventory Management in the Pharmaceutical Sector.” IEOM Society International, USA (2024). [\[Paper\]](#)

[C-2] **Md. Ferdaous Al-Farabe**, Ifte Kharul Islam Nahid, Md. Rafiul Islam, Md. Ratul Hasan Joy, Punno Chandra Saha, Shahed Mahmud, Md. Ariful Haque. “Advancing Retail Experience with Smart Shopping Carts: A Design Process Perspective for Bangladesh.” 6th International Conference on Mechanical, Industrial and Materials Engineering, Rajshahi, Bangladesh. [\[Paper\]](#)

Submitted

[S-1] **Md. Ferdaous Al-Farabe**, Md. Easir Rahat, Mhia Md. Zaglul Shahadat, Araf Masud, Md. Tajbir Hasan Shuvo. “ANN-Based Indoor Thermal Comfort Prediction and Monitoring System.” Submitted to the 29th International Conference on Computer and Information Technology (ICCIT 2026), IEEE Bangladesh Section. [\[link\]](#)

RESEARCH & WORK EXPERIENCE

Research Assistant

Aug 2026 – Present

Advisor: Md. Rasel Sarkar, Assistant Professor, Dept. of IPE, RUET

- Working on a proposal for a deep reinforcement learning-guided multi-objective metaheuristic for human-centric dynamic flexible job shop scheduling, building on a literature review of dynamic scheduling and RL-guided metaheuristics.

Industrial Trainee, Akij Bathware Limited

Mar 2025 – Apr 2025

AkijBashir Group of Industries, Mymensingh, Bangladesh

- Observed 5 production sections and found the grinding section to be the main bottleneck, caused by poor machine layout and scheduling. Used cycle time versus takt time analysis to propose a layout and scheduling improvement. Also designed a 5S audit sheet adopted plant-wide.

PROJECTS

Machine Learning & Optimization [GitHub]

Energy-Aware Production Optimizer

- Built a machine learning model that optimizes manufacturing yield and energy consumption together, reaching 98.3% F1 for yield prediction and 99.8% R² for energy forecasting, with Pareto trade-off analysis and an interactive dashboard.

Product Development

- Smart Shopping Cart [Link]
We designed an RFID-based smart cart that auto-detects items and shows the live bill along with the total cost on an LCD display, with a built-in anti-theft mechanism. I worked on the 3D design of the products, stress and thermal simulations, and the Arduino-based electronics.
- Conveyor Belt with ASRS [GitHub]
As a team, we built an Automated Storage and Retrieval System that sorts objects based on color and fault detection. My main contribution was on the sensor logic and calibration, along with writing the control code that made the system function smoothly.

Formula Student Projects

Team Crack Platoon, RUET

- Formula Bharat 2024: As Business Team Member, prepared Cost Report and Bill of Materials, applied Multi-Criteria Decision Making for vendor selection, and presented Concept Resource Management analysis. Business Plan Presentation placed 10th internationally. [Results]
- Formula Student Switzerland 2023: As Business Team Member, worked on sponsor management and contributed to business case presentation, which placed 4th (70/75). [Results]

Data Analytics [Tableau Profile]

- People-Watching Trends on Netflix
- Bangladesh Industry Sector Analysis
- Exploring Shopping Trends in USA [Kaggle Profile]

HONORS AND SCHOLARSHIPS

ISCEA Prize Global Case Competition	2025
Received a 70% scholarship in the ISCEA Prize Global Supply Chain Case Competition.	
2nd Runner-up, Poster Presentation, 8th RCF Career Fair	2023
Presented ECO CRACK, a concept for retrofitting diesel vehicles into electric ones.	
Technical Scholarship	2024
Awarded academic scholarship for three consecutive years.	
Talent Pool Board Scholarship	2016
Awarded by the Ministry of Education, Bangladesh.	

LEADERSHIP & ADVISORY ROLES

RUET IPE Club	Jun 2025 – Jun 2026
Secretary (Admin)	
- Organized and led major student events, including CRACKERJACK 2.0 (national business case competition with 100+ universities and 300 teams), Brainwave (intra-university case competition with 80+ participants), and Unilever’s Learn to Lead program, while mentoring juniors for Presentio 4.0 (presentation & public speaking competition for freshers) and growing the club’s LinkedIn engagement by 36%.	
Team Crack Platoon, RUET (Formula Student Team)	2023 – 2025
Head of Cost, Business Team	
- Led the cost department for Bangladesh’s first Formula Student team after joining as a Business Team member two years earlier.	
Greater Mymensingh Zilla Association, RUET	2025 – 2026
Vice President	
- Helped organize community events and guided new admission candidates from the greater Mymensingh district.	

SKILLS

Programming:	Python, C/C++, MATLAB, R
Machine Learning:	Deep Learning, TensorFlow, Keras, PyTorch, Scikit-learn
Optimization & Analytics:	Gurobi, Minitab, Power BI, Tableau
Design & Simulation:	AutoCAD, SolidWorks, Arduino
Languages:	Bengali (Native), Hindi, English

REFERENCES

Dr. Shahed Mahmud	Mohammad Rakib Raihan
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